# Full-Duplex Pilot Beam-Rectifer Array for Space Solar Power

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***Abstract*—With global energy consumption on the rise, Space Solar Power (SSP) remains a promising source of renewable energy. SSP systems employ pilot beams to ensure accurate target localization and phase control, with retrodirective power beaming serving as an efficient method of alignment. However, the pilot beam requires additional power overhead and space consumption. This work explores a 5.8 GHz full-duplex pilot-beam rectifying antenna (rectenna) array as a means of minimizing pilot power consumption and maximizing net power harvested. The full duplex system was realized via frequency separated pilot and power beams alongside an RF circulator to achieve isolation. The system reached peak RF-DC efficiency of 17.4% at 16 dBm of input power. A link analysis was conducted in MATLAB to simulate the expected net energy harvested. The results indicate a 80.18 megawatt peak in net energy harvested for the proposed system at pilot array radius of 10.28 m, which is 0.95% of the total power receiving aperture. At higher pilot array radii, increased hardware power overhead leads decreased power harvested.**

***Index Terms*—Space solar power, rectenna, full-duplex, retrodirective arrays, wireless power transfer, geostationary Earth orbit, link-budget analysis.**

## I. Introduction

Due to its ability to continuously harvest large amounts of energy, Space Solar Power (SSP) through radiative wireless power transfer (WPT) has emerged as a potential solution for growing energy demands. SSP satellites collect solar power and send energy to Earth via microwave power beaming: the use of electromagnetic beams to wirelessly transfer energy. Satellites in Geostationary Earth Orbit (GEO) have the benefit of constant power transfer.

Significant power loss can occur when the Earth rectifying antenna (rectenna) array and satellite are not in alignment, requiring the use of pilot beams [1]. In a pilot beam system, the Earth station sends a pilot signal that guides the satellite power beam array to align itself. However, this procedure requires precise phase control [2]. Many existing tracking algorithms, such as ones that use telemetry and tracking sensors, can be inefficient and costly [3]. Retrodirective arrays (RDAs) retransmit a signal in the direction in which it was received without knowledge of the said direction allowing for simplified beam control. Their purely analog circuitry eliminates computation delay [3] [4].

Despite efficiency benefits of RDAs, pilot beams require additional space and energy costs. These issues can be minimized through the use of a full-duplex pilot-rectifier array. A full-duplex system is capable of simultaneous transmission and reception. Thus, such systems are space efficient, as they do not require a dedicated pilot transmitting antenna. Additionally, arrays have higher gain than single antennas, decreasing the transmit power required for the pilot to be detected. Subsequently, the total energy efficiency of the WPT and the net power harvested (power harvested minus power spent) would increase. One notable downside of full-duplex systems is self-interference, a type of power loss which occurs when the transmitting and receiving signals interfere with each other [5]. This effect is minimized by using separate frequencies for the pilot transmission and power beaming, based on designs in previous literature [6].

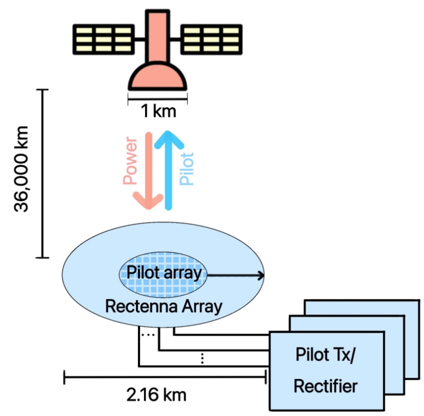


Fig. 1: Overview of the GEO SSP system with a full-duplex pilot beam-rectifer array.

Whereas most existing SSP research explores maximizing net power harvested through increased power beaming energy, there is little available on minimizing pilot beam expenditure [1]. Previous works in pilot power minimization have focused on low-power device applications over shorter ranges, but have not been investigated for long-range applications such as SSP [7] [8] [9].

This paper focuses on a proposed design and analysis of a 5.8 GHz full-duplex pilot-rectenna array in a GEO SSP system. A preliminary full-duplex pilot-rectenna element design was simulated using Ansys HFSS and Keysight ADS to determine the feasibility of the prototype full-duplex design in terms of its RF-DC efficiency. A MATLAB link analysis was then conducted to determine the required pilot transmission power, the power received at rectenna nodes, and the net energy harvested using investigated the pilot-rectenna design.

## II. System Overview

The GEO retrodirective SSP system under study is defined as having two parts: an uplink and a downlink. The uplink is defined as the pilot array's transmission towards the satellite's receiver for the purpose of target localization and power beam alignment. The antenna element used in the pilot-rectenna array consists of 4 dBi gain patch antennas in a circular configuration [10] [11]. The radius of the array is swept, with a maximum radius of 1080 meters. In full duplex operation, the pilot-rectenna array both transmits and receives; pilot beam transmission is done at 5.79GHz. Each element in the pilot-rectenna array is designed to be a node shown in Figure 1, which is a DC-combining energy harvesting scheme [12]. As seen in Figure 2, the rectennas have been connected to RF circulator based circuits in order to minimize interference during full duplex transmission and reception.

The downlink is defined as the satellite power beaming towards the pilot-rectenna array. The power beamed by the satellite was chosen as 500 megawatts, so the power reaching the pilot array would average 16 dBm, the input power which the designed pilot-rectenna is most efficient. The satellite's antenna array consists of 4 dBi gain patch antennas in a circular configuration, similar to the pilot array. The satellite array maintains a diameter of 1km based on designs in previous literature [13] [14], and beams down power at a microwave frequency of 5.8GHz, a trade off between minimizing aperture sizes and atmospheric losses [11].

## III. Preliminary Hardware Design

The proposed system realizes full-duplex operation via a frequency-separated architecture, eliminating the need for dual-polarized antennas and thereby simplifying the antenna design to a single patch element [6]. A three-port RF circulator is employed, wherein Port 1 is reciprocal and interfaces with the antenna to support concurrent transmit and receive signals, while Ports 2 and 3 are non-reciprocal, enforcing unidirectional signal flow for dedicated transmission and reception paths. Self-interference is mitigated through frequency separation, where the received power beam is at 5.8 GHz and the pilot transmission at 5.79 GHz. The power beam is routed through an impedance-matched network to avoid reflection and maximize power transfer to a full-wave single shunt rectifier, followed by a low-pass filter to suppress residual oscillations and stabilize the DC output prior to energy storage. The pilot is directed from a local oscillator to the antenna through the circulator. This architecture enables simultaneous transmission and reception on a single antenna through frequency separation and non-reciprocal signal routing, as shown in Fig. 2.

For the proposed architecture, a preliminary 5.8 GHz rectangular microstrip patch antenna was developed in ANSYS HFSS. Initial antenna dimensions were calculated from standard closed-form rectangular patch antenna design equations based on a Rogers RO4003C substrate ($\varepsilon_r = 3.38$, $h = 0.51$ mm). This yielded values of $W = 17.46$ mm and $L = 13.92$ mm. These dimensions were further optimized in HFSS to $W = 17.46$ mm and $L = 13.466$ mm. An SMA-fed port configuration was selected to support a modular framework.

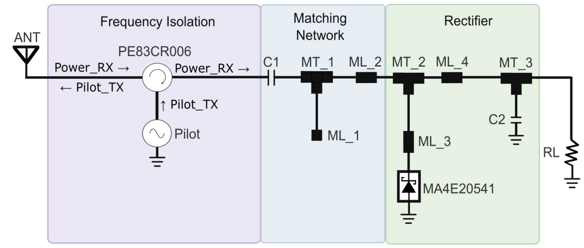


Fig. 2: Schematic of proposed pilot-rectifier architecture modeled in Keysight ADS, comprising circulator for frequency isolation, an impedance matching network, and a MA4E20541 full-wave shunt rectifier with load $R_L$, adapted from [12].

Simulation results showed that at 5.79 GHz, the normalized input impedance is approximately $1.0562 - j0.1892$, corresponding approximately 99.09% transmitted power, while at 5.80 GHz, the normalized input impedance is approximately $1.0979 - j0.0065$, corresponding to approximately 99.78% transmitted power. Fig. 3 illustrates the proposed frequency allocation, with 5.79 GHz designated for pilot beam transmission and 5.80 GHz for received power, both within the $-10$ dB bandwidth of the antenna. These results indicate effective impedance matching near the design frequency and support dual-frequency operation.

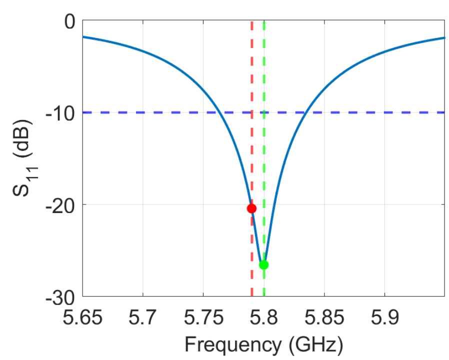


Fig. 3: $S_{11}$ return loss of a port-fed rectangular patch antenna with marked -10dB bandwidth (blue dashed), the pilot transmission frequency at 5.79 GHz (red), and the power beam frequency at 5.8 GHz (green).

Additionally, supporting circuitry was implemented in Keysight ADS, wherein a Pasternack PE83CR006 circulator was matched to a full-wave shunt rectifier with a Macom MA4E20541 Schottky diode adapted from [12]. The circulator routes the incoming power harvesting signal (Power_RX) to the rectifier while simultaneously transmitting the pilot signal (Pilot_TX) for full-duplex operation. To confirm this, the frequency spectrum was captured at each circulator port as shown in Fig. 4. The simultaneous presence of both frequencies at the TX_RX node confirms full-duplex operation.

Prior to full system integration, the rectifier was independently characterized to establish a baseline RF-to-DC

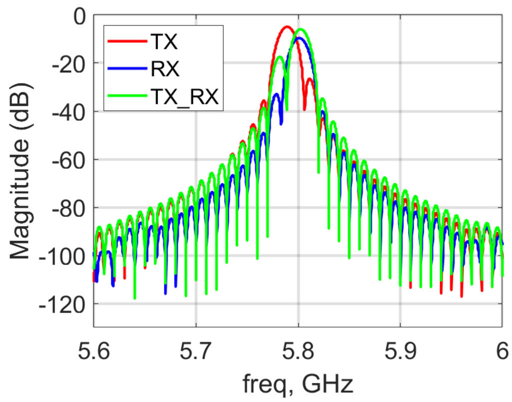


Fig. 4: Simulated frequency spectrum at each circulator port over 5.6–6.0 GHz, where the TX (red) and RX (blue) traces represent the isolated pilot transmit and power receive signals respectively, and the TX_RX (green) trace confirms the simultaneous presence of both signals, validating full-duplex operation.

conversion efficiency of 55% across load resistor $R_L$ at 13.26 dBm input power. Following integration of the Pasternack PE83CR006 circulator, the overall system peak efficiency shifted to 17.4% at 16 dBm input power, representing a 68.4% reduction in efficiency relative to the standalone rectifier baseline. Further investigation is warranted to identify the contributing factors to this efficiency degradation. Simulation results presented in Fig. 4 were obtained by sweeping the input RF power from -20 dBm to 35 dBm at a fixed pilot transmission power of 5 dBm. Additional simulations conducted at reduced pilot transmission power levels yielded negligible effects on RF-to-DC conversion efficiency, suggesting that pilot power is not a dominant contribution to loss. This motivates future investigation into potential RF power leakage at the circulator receive port into the pilot transmission port.

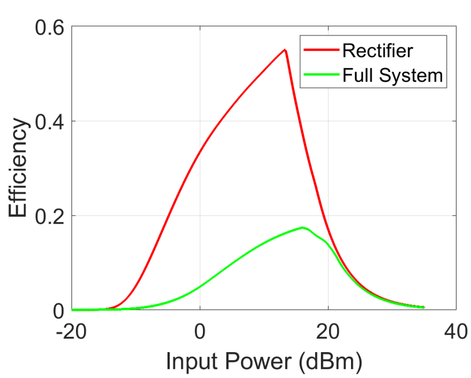


Fig. 5: RF-DC efficiency of the pilot-rectifier element.

## IV. Link Analysis

A link analysis is conducted to determine the performance of the proposed system. The net power harvested, $P_{net}$, was calculated by subtracting DC power consumption due to the pilot beam transmission, $P_{DC,pilot}$, from the energy sent down by the satellite, $P_{DC,EH}$, as seen in Equation 1.

$$P_{net} = P_{DC,EH} - P_{DC,pilot} \tag{1}$$

Assuming far field transmission, $P_{net}$ can be found using the Friis Free Space Equation on both the uplink and downlink, which is as follows:

$$P_R = G_R G_T P_T \frac{\lambda^2}{(4\pi R)^2} \tag{2}$$

where $P_R$ represents the RF power received, $G_R$ is the gain of the receiving antenna, $GT$ is the gain of the transmitting antenna, $P_T$ is the power being transmitted, and $R$ is the distance between the two antennas. On the pilot beam uplink, $P_R$ is a fixed constant, determined by the minimum power required for the beam to be detected by the receiving satellite antenna. According to the equation, it is possible to meet the same necessary $P_R$ with a lower $P_T$, or higher $R$, by bolstering the gain of the antennas. In this system, satellite receive antenna gain is held constant; therefore, this is accomplished by increasing pilot transmission gain. For this link analysis, focus remained on adjusting array dimensions to determine the gain and therefore net power harvested.

Under GEO orbit, $R$ is set to 36,000 km. The required uplink received power, $P_R$, was chosen to be -70dBm based on the Rx sensitivity of the phase conjugator [15], and satellite antenna array diameter was chosen to be 1 km. Patch antennas with 4 dBi of gain were chosen for the antenna elements on both the satellite and pilot-rectenna arrays with a spacing of $\lambda/2$ [16]. Both arrays receive on a singular element basis due to the retrodirective architecture and DC combining scheme [15]. Due to the large size of the studied arrays, a continuous circular aperture was used to approximate transmission gain for both uplink and downlink.

Continuous circular aperture gain is given by [16]:

$$G(\theta, \phi) = Du[\frac{2J_1(kasin(\theta))}{kasin(\theta)}]^2 \tag{3}$$

where $Du = \frac{4\pi}{\lambda^2}A$, $J_1$ is a Bessel function of the first order, $k = 2\pi/\lambda$, $\theta$ is elevation, $\phi$ is azimuth, $a$ is aperture radius, and $A$ is aperture area.

The satellite's antenna gain reaches its peak of 95.66 dBi at $\theta = 0$, and drops off at angles further from zero. The satellite's half-power beamwidth (HPBW) determined to be $6 * 10^{-5}$ degrees, which is when its gain drops to half of its peak gain. The energy harvesting aperture on earth was designed in alignment with the spot size of the HPBW, resulting in a radius of 1080 meters.

Using efficiencies simulated and shown in Figure 5, the RF power received at each Earth antenna was multiplied by the corresponding system efficiency and summed to find the total net DC power received.

In order to determine net power harvested, it was then necessary to subtract required transmit power from the power harvested as described in Equation 1. Different pilot array sizes were swept in order to illustrate the difference in required power for varying array sizes. The rectifier-only efficiency is used to calculate the power harvested from the areas outside of

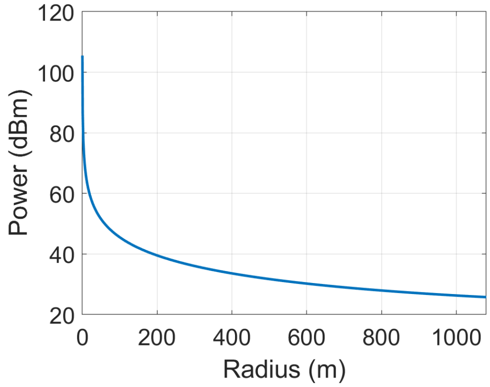


Fig. 6: Required pilot transmit power for different full-duplex pilot array sizes while the overall rectenna array is held constant at 1080m.

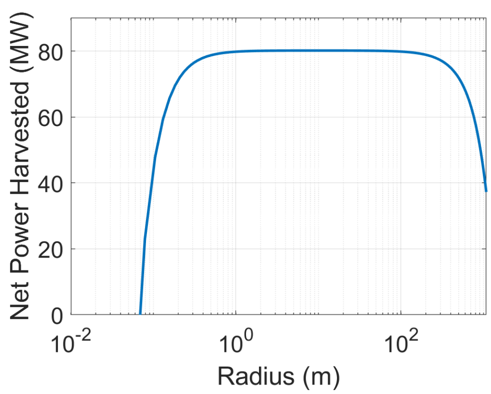


Fig. 7: Net power harvested for different full-duplex pilot array sizes while the overall rectenna array is held constant at 1080m.

the pilot array size. As seen in Figures 6 and 7, an increased pilot array size leads to sharp decrease in required transmit power, and therefore an increase in net harvested DC power when compared to smaller array sizes. This effect levels off quickly due to a trade off in lower RF-DC pilot-rectifier efficiency compared to rectifier-only efficiency, as seen in Figure 5. As a result, net power harvested peaks at 80.18 MW given a pilot-rectenna array radius of 10.28 meters which is 0.95% of the whole power receiving aperture.

## V. Conclusion

This work used link-budget analysis and MATLAB simulation to evaluate the feasibility of a prototype full-duplex pilot-beam rectenna array for a retrodirective GEO SPS application. Net harvested power was estimated using the simulated RF-DC efficiency of the proposed full-duplex pilot-rectifier element design. The analysis indicates that increased pilot-beam array gain in the proposed full-duplex framework can reduce the uplink transmit power required for effective pilot-beam transmission, although a larger pilot array creates an energy harvesting trade-off due to the proposed pilot-rectifier's lower RF-DC efficiency compared to just the rectifier's. These results suggest that full-duplex operation can improve both net power efficiency and available energy-harvesting area.

The present analysis assumes far-field transmission, which limits efficiency due to diffraction, and neglects multi-path effects; both limitations should be addressed in future work. Furthermore, the hardware design should be further developed in order to increase system efficiency.


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